



HWB-Net: A Novel High-Performance and Efficient Hybrid Waveform Bandwidth Extension Method

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Abstract

Speech Bandwidth Extension (BWE) reconstructs missing high-frequency components in narrowband signals to enhance perceptual quality. Conventional signal-processing methods face performance limitations, while deep-learning approaches require substantial computational resources. To address these issues, we present HWB-Net, a novel Hybrid Waveform Bandwidth network that combines half-wave rectified signals with raw waveforms through a lightweight architecture, with a lightweight architecture, reducing both complexity and cost. Specifically, integrating half-wave rectification into BAE-NET-Lite enhances the model's understanding of speech signals, leading to improved subjective and objective results, and replacing the decoder with a simple linear layer plus a weighted Gaussian mixture model (WGMM) significantly cuts down parameters (0.20M) and computational complexity (0.013G/s). Evaluations demonstrate HWB-Net's competitive performance and practical viability for real-time communication through balanced efficiency-accuracy tradeoffs.

Index Terms: bandwidth extension, half wave rectification, adaptive bandwidth extension, low-complexity, magnitude inpainting

1. Introduction

Bandwidth extension (BWE), a critical technology for improving speech quality in applications like telecommunications and voice assistants, has garnered significant research attention. Early signal-processing-based methods established foundational frameworks through various approaches: spectral envelope modeling (e.g., Linear Predictive Coding [1]), statistical mapping (e.g., Gaussian Mixture Models [2], Hidden Markov Models [3]), and nonlinear signal processing techniques including harmonic generation via octave extension [4], full/half-wave rectification [5], chaotic prediction-based reconstruction [6], and phase space reconstruction [7]. While these methods pioneered high-frequency reconstruction by exploiting low-band correlations, their reliance on statistical assumptions and simplified signal models often limited high-band reconstruction accuracy, particularly in complex acoustic environments.

Recent advances in deep learning have spawned diverse neural-network-based bandwidth extension (BWE) models, which primarily fall into two paradigms. Frequency-domain approaches [8, 9, 10, 11, 12, 13, 14] learn spectral mappings, exemplified by AERO's complex-spectrum recovery mechanism for simultaneous magnitude and phase reconstruction [12]. Time-domain methods [15, 16, 17, 18, 19] directly process waveforms through architectures like AudioUnet's CNNs [18],

TF-Net's hybrid time-frequency fusion [19], and SEANet's fully convolutional framework [17]. Despite their effectiveness, both paradigms suffer from prohibitive computational costs and model sizes, severely limiting practical deployment.

BAE-Net[13] features a dual-branch architecture comprising a Magnitude Inpainting Network (MI-Net) and a Phase Refinement Network (PR-Net). The MI-Net handles high-frequency magnitude inpainting for speech signals with variable bandwidth, while the PR-Net optimizes the flipped phase, enhanced by an information interaction module for cross-component coordination. Its lightweight variant, BAE-Net-lite, prioritizes magnitude inpainting using the flipped phase for waveform reconstruction. Despite lower computational demands compared to other models, BAE-Net-lite remains challenging to deploy on resource-constrained edge devices.

In machine learning frameworks such as BAE-Net and AP-BWE[20], high-frequency components in model inputs exhibit limited information density due to direct upsampling from low-sampling-rate speech signals. Naturally, it occurs to us that we can use some useful signals to fill this gap. Previously, a lot of work has been done to extrapolate high-frequency information through [5, 21, 22]. Among these, we adopt Half-Wave Rectification (HWR)[5], a foundational nonlinear operator that selectively retains the positive or negative half-cycle of the waveform. This choice is motivated by its algorithmic simplicity, computational efficiency, and interpretability, making it particularly suitable for preliminary exploration in resource-constrained scenarios.

Against this backdrop, we integrate traditional half-wave rectification into BAE-Net. By applying it to the input speech signal, we can access additional valuable information that existing methods have overlooked. This new-found information enriches the model's understanding of speech signal characteristics, leading to a more accurate speech signal reconstruction. Through experiments, this novel utilization of half-wave rectification has been proven to significantly improve the model's performance. Notably, it plays a crucial role in enhancing the details of high-frequency components, which are vital for achieving high-quality perceptual speech.

In addition, we have made substantial modifications to the decoder part of the network. Specifically, we replaced the original decoder with a simple linear layer combined with Weighted Gaussian Mixture Models (WGMM). This redesigned decoder not only effectively reduces the number of parameters but also significantly decreases the computational complexity. The reduction in parameters is of great significance as it allows the model to be more easily deployed on edge devices with limited computational resources. For example, in our experiments, the new decoder architecture has shown a significant reduction in memory usage and computational time compared to the original

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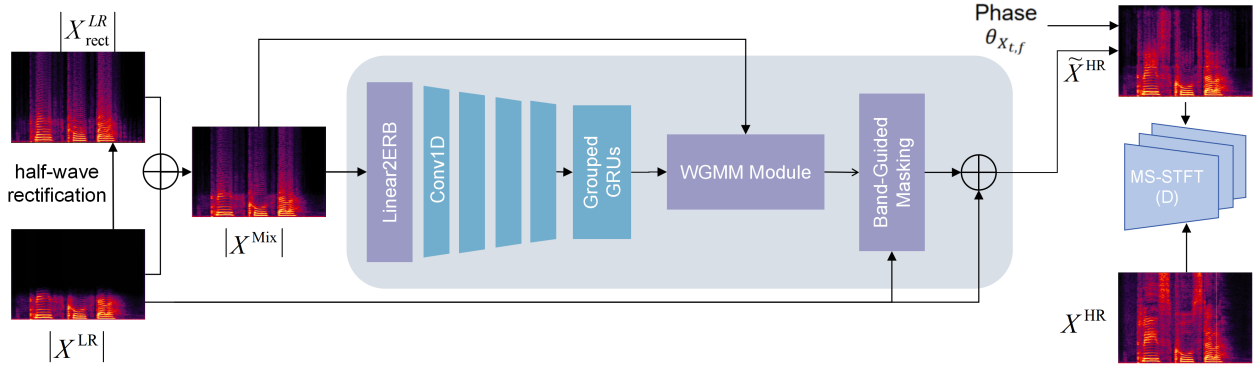


Figure 1: The flowchart of the proposed HWB-Net

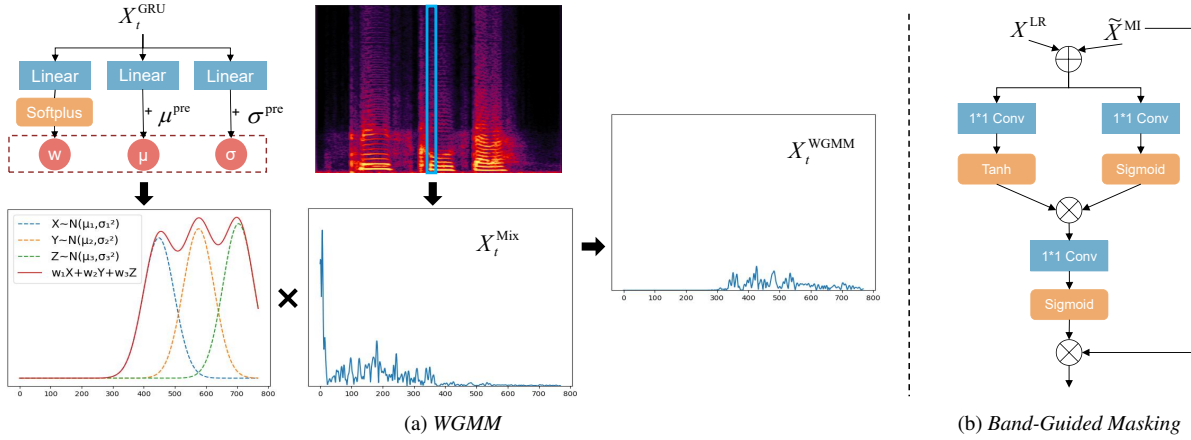


Figure 2: (a)Weighted Gaussian Mixture Models (WGMM) module. (b)Band-Guided Masking module.

design, making it a more viable option for real-time applications on mobile or other edge devices.

In summary, our proposed method has the following contributions:

- We introduce traditional half-wave rectification to enrich the input information and improve the model’s performance in reconstructing high-frequency components of speech signals.
- By redesigning the decoder with WGMM, we greatly reduce the computational complexity and parameter count, making the model more suitable for deployment on edge devices.
- Overall, our method overcomes the limitations of existing BWE solutions and provides a more efficient and effective alternative for speech bandwidth extension in practical RTC scenarios.

Through comprehensive experiments and evaluations, we will thoroughly demonstrate the superiority of our approach in terms of performance, computational efficiency, and its ability to adapt to variable bandwidth conditions.

2. Methodology

2.1. Overview

The overall diagram of the proposed system is illustrated in Figure 1, which is based on BAE-Net and incorporates two main improvements. First, we apply half-wave rectification to the

low-resolution (LR) input signal in the time domain to augment high-frequency information. Second, we replace the decoder of BAE-Net with a module based on the weighted Gaussian mixture model (WGMM), called the WGMM Module, to output the result of half-wave rectification multiplied by the Weighted Density Function (wdf).

2.2. Half-wave Rectification

We perform half-wave rectification on the input LR signal in the time domain to augment high-frequency information. The half-wave rectification operation is defined as:

$$X_{\text{rect}}^{LR} = \max(0, X^{LR}) \quad (1)$$

where X^{LR} represents the input LR signal.

Then, the magnitude spectra of the LR signal and the rectified signal are summed to form the actual input to the model, denoted as X^{Mix} .

2.3. WGMM Module

As illustrated in Figure 2a, our architecture processes the GRU output $X_t^{\text{GRU}} \in \mathbb{R}^{T \times F}$ through three parallel linear transformations to estimate Weighted Gaussian Mixture Model (WGMM) parameters. For each temporal frame t , these transformations produce N sets of parameters $\{\mathbf{w}_{t,k}, \boldsymbol{\mu}_{t,k}, \boldsymbol{\sigma}_{t,k}\}$ corresponding to N Gaussian components. Here, $k \in [1, N]$ and repre-

sents the k -th set of parameters. Three key characteristics distinguish our WGMM implementation from conventional GMM approaches:

1. **Constrained Weighting:** The predicted weights $w_t > 0$ are exempt from the typical unit-sum constraint, allowing for flexible component scaling.
2. **Frequency-Anchored Means:**
 - Base means μ^{pre} is a set of pre-set values that are strategically distributed across the target frequency range
 - Final means combine base values with learned offsets $\mu_{t,k}$: $\mu_{t,k} = \mu_k^{\text{pre}} + \mu_{t,k}$
 - Example: For $N = 4$ predicting 500-1000Hz range: $\mu^{\text{pre}} = \{600, 700, 800, 900\}$
3. **Constrained Variance Estimation:**
 - Base deviation σ^{pre} is a pre-set value that ensures numerical stability
 - Final deviations employ exponential rectification: $\sigma_{t,k} = \sigma^{\text{pre}} + \sigma_{t,k}$
 - *Implementation Note:* Through empirical validation, we found fixing $\sigma_{t,k} \equiv \sigma^{\text{pre}}$ achieved comparable performance to learnable variants while significantly improving training stability. This design choice reflects the minimal sensitivity of downstream tasks to precise variance values in our architecture.

The WGMM computes frame-wise spectral estimates through density composition:

$$\mathbf{f}_t(f) = \sum_{k=1}^N w_{t,k} \cdot \mathcal{N}(f \mid \mu_{t,k}, \sigma_{t,k}^2) \quad (2)$$

where $\mathbf{f}_t(f) \in \mathbb{R}^F$ represents the frequency distribution weights. These weights modulate the mixture spectrum through element-wise multiplication:

$$|X_t^{\text{WGMM}}| = \mathbf{f}_t \odot |X_t^{\text{Mix}}| \quad (3)$$

Here $|X_t^{\text{Mix}}|$ denotes the input mixture's amplitude spectrum, and $\odot$ represents the Hadamard product. This formulation enables frequency-specific gain adjustment guided by the WGMM's probabilistic structure while preserving phase information from the original mixture.

The predicted half-wave rectified output $|X^{\text{WGMM}}|$ and the input $|X^{LR}|$ are then fed into the Band-Guided Masking module, as illustrated in Figure 2b. The output from this module is combined with the input $|X^{LR}|$ to obtain the final magnitude spectrum prediction, also denoted as $|\hat{X}^{\text{MI}}|$.

Finally, during the training phase, the predicted speech $\hat{X}^{HR}$ is reconstructed by using the phase of the high-resolution speech X^{HR} and the final magnitude spectrum $|\hat{X}^{\text{MI}}|$. In contrast, during the inference phase, the predicted speech $\hat{X}^{HR}$ is reconstructed using the low-resolution input phase X^{LR} and the final magnitude spectrum $|\hat{X}^{\text{MI}}|$.

2.4. Training and Loss Function

Our model employs the same loss function and GAN structure as BAE-Net to ensure effective recovery of the HR magnitude spectrum and phase information. Specifically, the loss function includes waveform loss, multiresolution STFT loss, adversarial loss, and feature matching loss.

Waveform Loss: The waveform loss L_{wav} measures the difference between the generated waveform and the real waveform

to match the overall shape and the phase. **Multi-Resolution STFT Loss:** The multi-resolution STFT loss L_{stft} inspired by [23, 24, 25] is introduced to improve the perceptually-related auditory quality, which is a combination of the spectral convergence loss and the logarithmic magnitude spectral L1 loss with different FFT analysis parameters. In our experiments, the number of FFT bins is set to $\{512, 1024, 2048\}$, the hop length is set to $\{50, 120, 240\}$, and the window size is set to $\{240, 600, 1200\}$. **Adversarial Loss:** The adversarial loss L_{adv} is used to stabilize the competitive relationship between multiple discriminators D_i and the generator G , employing the relativistic average least-square loss (RaLSGAN) proposed in [26]. The RaLSGAN loss is defined as follows:

$$L_D^{\text{adv}} = \mathbb{E}_s [(D(s) - 1)^2] + \mathbb{E}_{\hat{s}} [(D(\hat{s}) + 1)^2] \quad (4)$$

$$L_G^{\text{adv}} = \mathbb{E}_{\hat{s}} [(D(\hat{s}) - 1)^2] + \mathbb{E}_s [(D(s) + 1)^2] \quad (5)$$

where $\hat{s}$ denotes the generated HR time-domain signal, and s denotes the target signal. **Feature Matching Loss:** The feature matching loss L_{feat} [27] is used to minimize the L1 distance between the feature maps of the discriminator's internal outputs for real HR speech and those for the corresponding generated speech.

The generator's overall loss is a weighted sum of these four losses and can be formulated as:

$$L_G = \lambda_{\text{wav}} L_{\text{wav}} + \lambda_{\text{stft}} L_{\text{stft}} + L_{\text{adv}} + \lambda_{\text{feat}} L_{\text{feat}} \quad (6)$$

The weight parameters are set as follows: $\lambda_{\text{wav}} = 100$, $\lambda_{\text{stft}} = 0.5$, $\lambda_{\text{feat}} = 10$

3. Dataset & Implementation

Our experimental setup is designed to address speech super-resolution tasks, inspired by the BAE-Net framework, with specific adaptations to meet our research objectives. We evaluated our approach using the widely recognized VCTK data set [28], which contains approximately 44 hours of clean speech sampled at 48 kHz from 110 speakers. For training, we utilize data from 108 speakers, reserving the remaining 2 speakers for testing.

3.1. Data Preprocessing

Our methodology specifically targets bandwidth extension under low-power edge computing constraints, where hardware limitations typically enforce low sampling rates (4-6 kHz range). This operational context necessitates particular attention to spectral recovery quality in sub-6kHz sampling scenarios.

Input speech signals undergo STFT preprocessing using a Hann window with 512-sample frame length and 256-sample shift. The 1536-point STFT configuration yields 769-dimensional spectral features, optimizing frequency resolution while maintaining temporal coherence. LR signals are generated from HR counterparts through a Butterworth low-pass filter (2-3 kHz cutoff, order:5-10). This processing chain creates matched LR-HR pairs where LR signals represent band-limited inputs and HR signals serve as reconstruction targets.

3.2. Training Details

The training process employs the Adam optimizer [29] with a learning rate of $2e-5$. The network architecture focuses on an

Table 1: Overall Performance Comparison

Method	Para. (M)	MACs (G/s)	LSD ↓	DNSMOS P.808 ↑	PESQ ↑	NISQA ↑
Sinc	-	-	3.26	2.90	3.11	2.85
AERO	19.41	35.78	0.90	3.29	2.84	3.54
BAE	2.83	0.128	0.89	3.34	2.79	3.61
BAE-Lite	0.57	0.035	0.89	3.27	2.77	3.74
BAE-Lite-Small	0.20	0.013	0.92	3.22	2.63	3.59
HWB	0.20	0.013	0.94	3.35	2.85	3.86

Table 2: Ablation Study Results

Method	Para. (M)	MACs (G/s)	LSD ↓	DNSMOS P.808 ↑	PESQ ↑	NISQA ↑
BAE-Lite	0.57	0.035	0.89	3.27	2.77	3.74
+ HWR	0.57	0.035	0.90	3.37	2.83	3.76
+ HWR & WGMM (HWB)	0.20	0.013	0.94	3.35	2.85	3.86

encoder designed to capture temporal dependencies and spectral features. The encoder includes four 1D convolutional layers (Layer 1: input channels 128, output channels 128; Layer 2: 128 input, 128 output; Layer 3: 128 input, 64 output; Layer 4: 64 input, 64 output), all with a kernel size of 3, followed by a Recurrent Neural Network (RNN) with input and hidden sizes of 64 to model temporal dynamics. The WGMM configuration employs $N = 64$ components with base parameters $\sigma^{\text{pre}} = 20$ and $\mu^{\text{pre}} \in [\frac{769}{4}, 769]$, where the mean bounds derive from the 1:4 LR-HR sampling rate ratio and 769-dimensional spectral features. This constrained initialization ensures stable distribution estimation during training.

3.3. Evaluation Metrics

The performance of the model is evaluated through multiple metrics. The Log-Spectral Distance (LSD) is employed to quantify spectral differences. For the objective and auditory assessment of speech quality, the PESQ metric [30] is utilized. Moreover, deep-learning-based metrics such as DNSMOS P.808 [31] and NISQA [32, 33] are adopted for a comprehensive evaluation of speech quality.

4. Results & Analysis

4.1. Overall Performance Comparison

Table 1 demonstrates HWB-Net’s state-of-the-art efficiency-quality tradeoffs. With merely 0.20M parameters, our model achieves 3.35 DNSMOS score, outperforming the 19.41M-parameter AERO system by 1.8% while reducing computational cost to 0.013G MACs (37.1% of BAE-Lite). The PESQ improvement of 2.2% over BAE-Lite-Small (2.85 vs 2.63) combined with superior NISQA performance (3.86 vs 3.59 in comparable configurations) confirms the effectiveness of our hybrid waveform bandwidth extension design.

4.2. Component Ablation Study

As quantified in Table 2, the half-wave rectification (HWR) module improves the DNSMOS score by 3.06% (3.27→3.37) and slightly enhances NISQA (3.74→3.76) without additional computational cost. Subsequent integration of the softplus-constrained WGMM achieves dual benefits: a 64.91% parameter reduction (0.57M→0.20M) with minimal quality degradation (DNSMOS 3.37→3.35), while the 5.62% LSD increase

(0.89→0.94) reflects our systematic design trade-off. Notably, this architectural refinement simultaneously elevates NISQA by 2.66% (3.76→3.86), demonstrating effective perceptual quality improvement despite spectral matching constraints. The WGMM architecture’s softplus-constrained linear superposition inherently limits exact spectral matching but implements perceptually-guided component selection through: (1) non-negative weighting for noise floor suppression via attenuated low-weight components, and (2) adaptive emphasis through relative weight magnitudes that theoretically enforce energy concentration in dominant spectral regions. This framework maintains robust quality metrics with PESQ at 2.85 versus baseline 2.83 and achieves superior NISQA performance (3.86 vs. 3.59-3.74 in comparable models), demonstrating effective balance between model efficiency and perceptual preservation.

5. Conclusions & Future Work

Our work advances speech bandwidth extension through two key innovations: (1) The novel integration of half-wave rectification in hybrid waveform models achieves measurable quality enhancement without computational overhead, establishing HWR as an efficient perceptual optimization mechanism; (2) The proposed WGMM architecture enables unprecedented model efficiency (0.2M parameters, 0.013G MACs) through strategic component selection while maintaining perceptual fidelity. This dual-mechanism framework demonstrates systematic prioritization of perceptual quality over spectral matching precision, confirming its viability for resource-constrained real-time applications. The balanced quality-efficiency tradeoffs set a new benchmark for practical bandwidth extension systems.

Three promising directions emerge from this research: (1) Hybrid signal processing integrating nonlinear high-frequency extrapolation techniques with data-driven models to enhance spectral diversity; (2) Distributional refinement through task-specific mixture components, potentially replacing Gaussian bases with heavy-tailed distributions (Laplacian/Exponential mixtures) better suited for vocal tract excitation patterns; (3) multi-scale probabilistic architectures using parallel Gaussian mixtures with varied initial parameters $\{\mu_k, \sigma_k^2\} = \{1, 1\}, \{2, 2\}, \dots$ through $p(x) = \sum w_k \mathcal{N}(x; \mu_k, \sigma_k^2)$.

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